

Hardware Commands vs. Data Arrays

Why **w** descriptors are fundamentally different from rigid **x** descriptors.

❌ The Old Way: **x** Descriptors (Rigid Hardware Commands)

```
</> index.html  
srcset="img.jpg 2x"  
Directive: If the screen DPR is 2,  
you must use this image.
```

📦 What the browser knows

img.jpg
Width: Unknown
Target: 2x Displays



⚠️ The Flaw

The browser ignores CSS layout.
A 2x screen always downloads
the 2x image, even if the image
is displayed at a tiny 50px width!

VS

✅ The Modern Way: **w** Descriptors (Flexible Asset Inventories)

```
</> index.html  
srcset="sm.jpg 400w, md.jpg 800w"  
Data: purely documents the native pixel widths  
of available assets.
```

🖼️ Array of Available Assets

small.jpg
400w
Native Width:
400px

med.jpg
800w
Native Width:
800px

lg.jpg
1200w
Native Width:
1200px

Layout Width: 50px
× Screen DPR: 2
= 100px Needed



✅ Optimal
Chooses 400w!
(Smallest ≥ 100px)

💡 The Solution

The **w** descriptor dictates nothing.
It just lists facts. The browser de-
cides which asset to use based on
Layout Width AND Screen Density.